

Package ‘rENM.reports’

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Type Package

Title Report generation tools for the rENM Framework

Version 0.1.0

Description Provides report generation tools for the rENM Framework, including compilation of summary tables, assembly of per-topic report pages, and production of the final species report by combining PDF products from across the rENM pipeline.

Depends R (>= 4.1.0)

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URL <https://github.com/rENM-Framework/rENM.reports>

BugReports <https://github.com/rENM-Framework/rENM.reports/issues>

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dplyr,
graphics,
grDevices,
grid,
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magick,
officer,
openxlsx,
pdftools,
png,
readr,
stats,
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Suggests chromote,
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assemble_centroid_trends_page
Assemble a "Centroid Trends" page

Description

Builds a single-page PDF combining a suitability trend map with centroids, a centroid trend summary table, and a caption into a structured layout.

Usage

```
assemble_centroid_trends_page(
    alpha_code,
    page_width_in = 8.5,
    page_height_in = 11,
    dpi = 300,
    margin_left_in = 0.75,
    margin_right_in = 0.75,
    margin_bottom_in = 0.75,
    margin_top_in = 1,
    v_gap_in = 0.3,
    v_caption_gap_in = 0.3,
    map_width_factor = 1,
    table_width_factor = 0.75,
    table_max_height_in = 2.5,
    table_pdf_dpi = 600L,
    trim_map = TRUE,
    trim_table = TRUE,
    trim_fuzz = 8,
    caption_dpi = 600L,
    debug_frames = FALSE
)
```

Arguments

alpha_code	Character. Species alpha code (case-insensitive).
page_width_in	Numeric. Page width in inches. Default 8.5.
page_height_in	Numeric. Page height in inches. Default 11.
dpi	Numeric. DPI for raster composition. Default 300.
margin_left_in	Numeric. Left margin in inches. Default 0.75.
margin_right_in	Numeric. Right margin in inches. Default 0.75.
margin_bottom_in	Numeric. Bottom margin in inches. Default 0.75.
margin_top_in	Numeric. Top margin in inches. Default 1.00.
v_gap_in	Numeric. Vertical gap between map and table in inches. Default 0.30.
v_caption_gap_in	Numeric. Vertical gap between table and caption in inches. Default 0.30.
map_width_factor	Numeric. Fraction (0 to 1) of content width used by the map. Default 1.00.
table_width_factor	Numeric. Fraction (0 to 1) of content width used by the table. Default 0.75.
table_max_height_in	Numeric. Maximum table height in inches. Default 2.50.
table_pdf_dpi	Integer. DPI used when rasterizing the table PDF. Default 600.
trim_map	Logical. If TRUE, trim borders from the map PNG. Default TRUE.
trim_table	Logical. If TRUE, trim borders from the table image. Default TRUE.
trim_fuzz	Integer. Tolerance (0 to 100 percent) for trimming. Default 8.
caption_dpi	Integer. DPI used when rasterizing the caption PDF. Default 600.
debug_frames	Logical. If TRUE, draw thin frames around placed images for debugging. Default FALSE.

Details

Pipeline context This function composes a precomputed centroid trend map, a summary table, and a package caption into a single standardized reporting page. It operates entirely within the rENM project directory and does not rely on external scripts.

Inputs Inputs must exist at the following locations:

```
<project_dir>/runs/<alpha_code>/Trends/suitability/<alpha_code>-Suitability-Trend-With-Centroid-Trend-Map.png
<project_dir>/runs/<alpha_code>/Summaries/tables/<alpha_code>-Centroid-Trend-Summary.pdf
```

The caption is read from the installed package resource:

```
inst/captions/centroid_shift_caption.pdf
```

Outputs The function writes a single-page PDF to:

```
<project_dir>/runs/<alpha_code>/Summaries/pages/<alpha_code>-Centroid-Trends.pdf
```

Layout and composition

- The centroid trend map is placed at the top of the content area, scaled to fill the content width (subject to `map_width_factor`) while preserving its aspect ratio.
- A summary table is placed `v_gap_in` inches below the map, centered horizontally (subject to `table_width_factor`).
- A caption is placed `v_caption_gap_in` inches below the table, centered horizontally and never stretched (aspect ratio preserved).
- Layout dimensions are controlled in inches and converted to pixels using the specified DPI.

Caption rendering The caption PDF (`centroid_shift_caption.pdf`) is rasterized via `pdftools::pdf_convert()` at `caption_dpi` (default 600) to preserve original typography (including bold). It is trimmed to remove surrounding page whitespace, then centered horizontally and scaled down only if wider than the content area; otherwise it is drawn at native size.

Table rendering The table PDF is rasterized via `pdftools::pdf_convert()` at `table_pdf_dpi` (default 600) to improve text clarity. The table is then fitted within `table_width_factor` of the content width subject to `table_max_height_in`.

Trimming

- Uniform borders are optionally trimmed from the map (controlled by `trim_map`) and the table (controlled by `trim_table`) to remove hidden padding.
- The caption is always trimmed after rasterization to remove the surrounding page whitespace, so only the text block contributes to the layout height calculation.
- Trimming uses a tolerance defined by `trim_fuzz`.

Methods Image reading, trimming, scaling, and composition are handled by the `magick` package. The table and caption PDFs are rasterized by `pdftools`. Aspect-preserving resizing ensures consistent layout while respecting maximum width and height constraints.

Data requirements All required input files must exist prior to execution. The function will terminate with an error if any required input or the caption PDF is missing.

Value

Character. Invisibly returns the output PDF file path.

Side effects:

- Writes a PDF file to the project directory.
- Creates output directories if they do not exist.
- Emits a console message indicating the output path.

Examples

```
## Not run:
assemble_centroid_trends_page("CASP")

## End(Not run)
```

assemble_final_report *Assemble a final report for an rENM run*

Description

Combines a predefined or user-specified set of per-page PDF products into a single report PDF for a given species run. Optionally rasterizes the pages and draws centered page numbers on all but the cover page. Optionally prepends a front matter PDF and/or appends an appendix PDF sourced from rENM.reports.

Usage

```
assemble_final_report(
  alpha_code,
  pages = NULL,
  front_matter = NULL,
  appendix = "variables.pdf",
  page_numbers = TRUE,
  dpi = 150,
  cleanup = TRUE,
  verbose = TRUE
)
```

Arguments

alpha_code	Character. Four-letter species alpha code.
pages	Character vector. Optional. Specifies which pages to include and their order. May be either: • full filenames (e.g., "BETH-Suitability-Trends.pdf"), or • short keys (see Details; e.g., "Suitability-Trends") If NULL (default), the standard page set and order are used.
front_matter	Character. Optional. Filename of a PDF located in <code>rENM.reports::inst/resources/</code> to prepend.
appendix	Character. Optional. Filename of a PDF located in <code>rENM.reports::inst/resources/</code> to append.
page_numbers	Logical. Add page numbers (default TRUE).
dpi	Numeric. Rasterization resolution when numbering.
cleanup	Logical. Remove temporary files.
verbose	Logical. Print progress messages.

Details

By default, a fixed set of pages is assembled in a predefined order. Users may override this behavior via the `pages` argument to:

- reorder pages
- omit pages
- include a subset of pages

Optional front matter and appendix PDFs must be located in `inst/resources/` within the `rENM.reports` package.

Inputs

- Pre-generated page PDFs in `<rENM_project_dir()>/runs/<alpha_code>/Summaries/pages/`
- Optional front matter / appendix PDFs in `rENM.reports::inst/resources/`

Outputs

- Final report: `<rENM_project_dir()>/runs/<alpha_code>/Summaries/ <alpha_code>-Final-Report.pdf`
- Processing summary appended to `<rENM_project_dir()>/runs/<alpha_code>/_log.txt`

Value

Character. Invisibly returns the final PDF path.

Examples

```
## Not run:
assemble_final_report("BETH")

assemble_final_report(
  "BETH",
  front_matter = "Front-Matter.pdf",
  appendix = "Appendix.pdf"
)

## End(Not run)
```

`assemble_range_timeseries_page`

Assemble a "Range Time Series" page

Description

Builds a single-page PDF that places a range contact-sheet map above a caption in a structured layout.

Usage

```
assemble_range_timeseries_page(
  alpha_code,
  page_width_in = 8.5,
  page_height_in = 11,
  dpi = 300,
  margin_left_in = 0.75,
  margin_right_in = 0.75,
  margin_bottom_in = 0.75,
  margin_top_in = 1,
  v_gap_in = 0.3,
  map_width_factor = 1,
```

```

    trim_map = TRUE,
    trim_fuzz = 8,
    caption_dpi = 600L,
    debug_frames = FALSE
)

```

Arguments

<code>alpha_code</code>	Character. Species alpha code (case-insensitive).
<code>page_width_in</code>	Numeric. Page width in inches. Default 8.5.
<code>page_height_in</code>	Numeric. Page height in inches. Default 11.
<code>dpi</code>	Numeric. DPI for raster composition. Default 300.
<code>margin_left_in</code>	Numeric. Left margin in inches. Default 0.75.
<code>margin_right_in</code>	Numeric. Right margin in inches. Default 0.75.
<code>margin_bottom_in</code>	Numeric. Bottom margin in inches. Default 0.75.
<code>margin_top_in</code>	Numeric. Top margin in inches. Default 1.00.
<code>v_gap_in</code>	Numeric. Vertical gap between map and caption in inches. Default 0.30.
<code>map_width_factor</code>	Numeric. Fraction (0 to 1) of content width used by the map. Default 1.00.
<code>trim_map</code>	Logical. If TRUE, trim borders from the map PNG. Default TRUE.
<code>trim_fuzz</code>	Integer. Tolerance (0 to 100 percent) for trimming. Default 8.
<code>caption_dpi</code>	Integer. DPI used when rasterizing the caption PDF. Default 600.
<code>debug_frames</code>	Logical. If TRUE, draw thin frames around placed images for debugging. Default FALSE.

Details

Pipeline context This function composes a precomputed range contact-sheet image and a package caption into a single standardized reporting page. It operates entirely within the rENM project directory and does not rely on external scripts.

Inputs Inputs must exist at the following locations:

```
<project_dir>/runs/<alpha_code>/Summaries/maps/<alpha_code>-Range-Maps.png
```

The caption is read from the installed package resource:

```
inst/captions/range_timeseries_caption.pdf
```

Outputs The function writes a single-page PDF to:

```
<project_dir>/runs/<alpha_code>/Summaries/pages/<alpha_code>-Range-TimeSeries.pdf
```

Layout and composition

- The contact-sheet map is placed at the top of the content area, scaled to fill the content width (subject to `map_width_factor`) while preserving its aspect ratio.
- A caption is placed `v_gap_in` inches below the map, centered horizontally and never stretched (aspect ratio preserved).

- Layout dimensions are controlled in inches and converted to pixels using the specified DPI.

Caption rendering The caption PDF (range_timeseries_caption.pdf) is rasterized via `pdftools::pdf_convert()` at `caption_dpi` (default 600) to preserve original typography (including bold). It is trimmed to remove surrounding page whitespace, then centered horizontally and scaled down only if wider than the content area; otherwise it is drawn at native size.

Trimming

- Uniform borders are optionally trimmed from the map (controlled by `trim_map`) to remove hidden padding.
- The caption is always trimmed after rasterization to remove the surrounding page whitespace, so only the text block contributes to the layout height calculation.
- Trimming uses a tolerance defined by `trim_fuzz`.

Methods Image reading, trimming, scaling, and composition are handled by the `magick` package. The caption PDF is rasterized by `pdftools`. Aspect-preserving resizing ensures consistent layout while respecting maximum width and height constraints.

Data requirements All required input files must exist prior to execution. The function will terminate with an error if the map PNG or the caption PDF are missing.

Value

Character. Invisibly returns the output PDF file path.

Side effects:

- Writes a PDF file to the project directory.
- Creates output directories if they do not exist.
- Emits a console message indicating the output path.

Examples

```
## Not run:
assemble_range_timeseries_page("CASP")

## End(Not run)
```

```
assemble_state_trends_page
```

Assemble a "State Trends" page

Description

Builds a single-page PDF that combines a state-level suitability map and a state hot spot map with a summary table and caption into a structured layout.

Usage

```

assemble_state_trends_page(
  alpha_code,
  page_width_in = 8.5,
  page_height_in = 11,
  dpi = 300,
  margin_left_in = 0.75,
  margin_right_in = 0.75,
  margin_bottom_in = 0.75,
  margin_top_in = 1,
  h_gutter_in = 0.2,
  v_gap_in = 0.3,
  map_width_factor = 1,
  table_width_factor = 0.75,
  table_max_height_in = 2.5,
  prefer_table_pdf = TRUE,
  table_pdf_dpi = 600,
  table_png_supersample = 1.5,
  table_sharpen = TRUE,
  trim_maps = TRUE,
  trim_table = TRUE,
  trim_fuzz = 8,
  caption_dpi = 600L,
  v_caption_gap_in = 0.3,
  append_version_to_filename = FALSE,
  debug_frames = FALSE
)

```

Arguments

<code>alpha_code</code>	Character. Species code (case-insensitive).
<code>page_width_in</code>	Numeric. Page width in inches. Default 8.5.
<code>page_height_in</code>	Numeric. Page height in inches. Default 11.
<code>dpi</code>	Numeric. DPI for raster composition. Default 300.
<code>margin_left_in</code>	Numeric. Left margin in inches. Default 0.75.
<code>margin_right_in</code>	Numeric. Right margin in inches. Default 0.75.
<code>margin_bottom_in</code>	Numeric. Bottom margin in inches. Default 0.75.
<code>margin_top_in</code>	Numeric. Top margin in inches. Default 1.00.
<code>h_gutter_in</code>	Numeric. Horizontal gap between maps in inches. Default 0.20.
<code>v_gap_in</code>	Numeric. Vertical gap between maps and table in inches. Default 0.30.
<code>map_width_factor</code>	Numeric. Fraction (0 to 1) of content width used by the map row. Default 1.00.
<code>table_width_factor</code>	Numeric. Fraction (0 to 1) of content width used by the table. Default 0.75.
<code>table_max_height_in</code>	Numeric. Maximum table height in inches. Default 2.50.

prefer_table_pdf	Logical. If TRUE and a PDF exists, read the table using <code>magick::image_read_pdf()</code> . Default TRUE.
table_pdf_dpi	Integer. DPI used when rasterizing table PDF. Default 600.
table_png_supersample	Numeric. Supersampling factor for PNG tables; values greater than or equal to 1 increase resolution (for example, 2 = 200 percent). Default 1.5.
table_sharpen	Logical. If TRUE, apply mild sharpening after scaling. Default TRUE.
trim_maps	Logical. If TRUE, trim borders from map PNGs. Default TRUE.
trim_table	Logical. If TRUE, trim borders from the table image. Default TRUE.
trim_fuzz	Integer. Tolerance (0 to 100 percent) for trimming. Default 8.
caption_dpi	Integer. DPI used when rasterizing the caption PDF. Default 600.
v_caption_gap_in	Numeric. Vertical gap in inches between the table and the caption. Default 0.30.
append_version_to_filename	Logical. If TRUE, append a timestamp to the output filename to avoid preview caching. Default FALSE.
debug_frames	Logical. If TRUE, draw thin frames around placed images for debugging. Default FALSE.

Details

Pipeline context This function composes multiple precomputed visualization products into a single standardized reporting page. It operates entirely within the rENM project directory and does not rely on external scripts.

Inputs Inputs must exist at the following locations:

```
<project_dir>/runs/<alpha_code>/Trends/suitability/<alpha_code>-Suitability-Trend-State-Analysis
<project_dir>/runs/<alpha_code>/Trends/suitability/<alpha_code>-Suitability-Trend-State-Analysis
<project_dir>/runs/<alpha_code>/Summaries/tables/<alpha_code>-Suitability-Trend-Summary.png
(Optional) PDF table:
<project_dir>/runs/<alpha_code>/Summaries/tables/<alpha_code>-Suitability-Trend-Summary.pdf
```

The caption is read from the installed package resource:

```
inst/captions/hotspot_summary_caption.pdf
```

Outputs The function writes a single-page PDF to:

```
<project_dir>/runs/<alpha_code>/Summaries/pages/<alpha_code>-State-Trends.pdf
```

Layout and composition

- Two top-aligned maps (State Analysis and Hotspots) are placed under a configurable top margin.
- The map row can be narrowed and centered using `map_width_factor`.
- A summary table is placed exactly `v_gap_in` inches below the taller of the two maps.
- The table can be narrowed and centered using `table_width_factor`.
- A caption is placed `v_caption_gap_in` inches below the table, centered horizontally and never stretched (aspect ratio preserved).

- Layout dimensions are controlled in inches and converted to pixels using the specified DPI.

Caption rendering The caption PDF (hotspot_summary_caption.pdf) is rasterized via `pdftools::pdf_convert()` at `caption_dpi` (default 600) to preserve original typography (including bold). It is centered horizontally and scaled down only if wider than the content area; otherwise it is drawn at native size.

Table rendering

- If `prefer_table_pdf = TRUE`, the table is read from PDF using `magick::image_read_pdf()` and rasterized at `table_pdf_dpi` (for example, 600) to improve text clarity.
- If using a PNG, `table_png_supersample` (for example, 2 for 200 percent) increases resolution prior to fitting.
- Optional sharpening can be applied using `table_sharpen = TRUE`.

Trimming

- Uniform borders are trimmed from maps and optionally from the table to remove hidden padding.
- The caption is always trimmed after rasterization to remove the surrounding page whitespace, so only the text block contributes to the layout height calculation.
- Trimming uses a tolerance defined by `trim_fuzz`.

Methods Image reading, trimming, scaling, and composition are handled by the `magick` package. The caption PDF is rasterized by `pdftools`. Aspect-preserving resizing ensures consistent layout while respecting maximum width and height constraints.

Data requirements All required input files must exist prior to execution. The function will terminate with an error if required PNGs, table sources, or the caption PDF are missing.

Value

Character. Invisibly returns the output PDF file path.

Side effects:

- Writes a PDF file to the project directory.
- Creates output directories if they do not exist.
- Emits a console message indicating the output path.

Examples

```
## Not run:
assemble_state_trends_page("CASP")

## End(Not run)
```

assemble_suitability_timeseries_page

Assemble a "Suitability Time Series" page

Description

Builds a single-page PDF that places a suitability contact-sheet map above a caption in a structured layout.

Usage

```
assemble_suitability_timeseries_page(
    alpha_code,
    page_width_in = 8.5,
    page_height_in = 11,
    dpi = 300,
    margin_left_in = 0.75,
    margin_right_in = 0.75,
    margin_bottom_in = 0.75,
    margin_top_in = 1,
    v_gap_in = 0.3,
    map_width_factor = 1,
    trim_map = TRUE,
    trim_fuzz = 8,
    caption_dpi = 600L,
    debug_frames = FALSE
)
```

Arguments

alpha_code	Character. Species alpha code (case-insensitive).
page_width_in	Numeric. Page width in inches. Default 8.5.
page_height_in	Numeric. Page height in inches. Default 11.
dpi	Numeric. DPI for raster composition. Default 300.
margin_left_in	Numeric. Left margin in inches. Default 0.75.
margin_right_in	Numeric. Right margin in inches. Default 0.75.
margin_bottom_in	Numeric. Bottom margin in inches. Default 0.75.
margin_top_in	Numeric. Top margin in inches. Default 1.00.
v_gap_in	Numeric. Vertical gap between map and caption in inches. Default 0.30.
map_width_factor	Numeric. Fraction (0 to 1) of content width used by the map. Default 1.00.
trim_map	Logical. If TRUE, trim borders from the map PNG. Default TRUE.
trim_fuzz	Integer. Tolerance (0 to 100 percent) for trimming. Default 8.
caption_dpi	Integer. DPI used when rasterizing the caption PDF. Default 600.
debug_frames	Logical. If TRUE, draw thin frames around placed images for debugging. Default FALSE.

Details

Pipeline context This function composes a precomputed suitability contact-sheet image and a package caption into a single standardized reporting page. It operates entirely within the rENM project directory and does not rely on external scripts.

Inputs Inputs must exist at the following locations:

```
<project_dir>/runs/<alpha_code>/Summaries/maps/<alpha_code>-Suitability-Maps.png
```

The caption is read from the installed package resource:

```
inst/captions/suitability_timeseries_caption.pdf
```

Outputs The function writes a single-page PDF to:

```
<project_dir>/runs/<alpha_code>/Summaries/pages/<alpha_code>-Suitability-TimeSeries.pdf
```

Layout and composition

- The contact-sheet map is placed at the top of the content area, scaled to fill the content width (subject to `map_width_factor`) while preserving its aspect ratio.
- A caption is placed `v_gap_in` inches below the map, centered horizontally and never stretched (aspect ratio preserved).
- Layout dimensions are controlled in inches and converted to pixels using the specified DPI.

Caption rendering The caption PDF (`suitability_timeseries_caption.pdf`) is rasterized via `pdftools::pdf_convert()` at `caption_dpi` (default 600) to preserve original typography (including bold). It is trimmed to remove surrounding page whitespace, then centered horizontally and scaled down only if wider than the content area; otherwise it is drawn at native size.

Trimming

- Uniform borders are optionally trimmed from the map (controlled by `trim_map`) to remove hidden padding.
- The caption is always trimmed after rasterization to remove the surrounding page whitespace, so only the text block contributes to the layout height calculation.
- Trimming uses a tolerance defined by `trim_fuzz`.

Methods Image reading, trimming, scaling, and composition are handled by the magick package. The caption PDF is rasterized by `pdftools`. Aspect-preserving resizing ensures consistent layout while respecting maximum width and height constraints.

Data requirements All required input files must exist prior to execution. The function will terminate with an error if the map PNG or the caption PDF are missing.

Value

Character. Invisibly returns the output PDF file path.

Side effects:

- Writes a PDF file to the project directory.
- Creates output directories if they do not exist.
- Emits a console message indicating the output path.

Examples

```
## Not run:
assemble_suitability_timeseries_page("CASP")

## End(Not run)
```

```
assemble_suitability_trends_page
```

Assemble a "Suitability Trends" page

Description

Builds a single-page PDF that combines a suitability trend map and a suitability change trend map with a caption into a structured layout.

Usage

```
assemble_suitability_trends_page(
  alpha_code,
  page_width_in = 8.5,
  page_height_in = 11,
  dpi = 300,
  margin_left_in = 0.75,
  margin_right_in = 0.75,
  margin_bottom_in = 0.75,
  margin_top_in = 1,
  h_gutter_in = 0.2,
  v_gap_in = 0.3,
  map_width_factor = 1,
  trim_maps = TRUE,
  trim_fuzz = 8,
  caption_dpi = 600L,
  debug_frames = FALSE
)
```

Arguments

<code>alpha_code</code>	Character. Species alpha code (case-insensitive).
<code>page_width_in</code>	Numeric. Page width in inches. Default 8.5.
<code>page_height_in</code>	Numeric. Page height in inches. Default 11.
<code>dpi</code>	Numeric. DPI for raster composition. Default 300.
<code>margin_left_in</code>	Numeric. Left margin in inches. Default 0.75.
<code>margin_right_in</code>	Numeric. Right margin in inches. Default 0.75.
<code>margin_bottom_in</code>	Numeric. Bottom margin in inches. Default 0.75.
<code>margin_top_in</code>	Numeric. Top margin in inches. Default 1.00.
<code>h_gutter_in</code>	Numeric. Horizontal gap between maps in inches. Default 0.20.

v_gap_in	Numeric. Vertical gap between maps and caption in inches. Default 0.30.
map_width_factor	Numeric. Fraction (0 to 1) of content width used by the map row. Default 1.00.
trim_maps	Logical. If TRUE, trim borders from map PNGs. Default TRUE.
trim_fuzz	Integer. Tolerance (0 to 100 percent) for trimming. Default 8.
caption_dpi	Integer. DPI used when rasterizing the caption PDF. Default 600.
debug_frames	Logical. If TRUE, draw thin frames around placed images for debugging. Default FALSE.

Details

Pipeline context This function composes two precomputed trend maps and a package caption into a single standardized reporting page. It operates entirely within the rENM project directory and does not rely on external scripts.

Inputs Inputs must exist at the following locations:

```
<project_dir>/runs/<alpha_code>/Trends/suitability/<alpha_code>-Suitability-Trend.png
<project_dir>/runs/<alpha_code>/Trends/suitability/<alpha_code>-Suitability-Change-Trend.png
```

The caption is read from the installed package resource:

```
inst/captions/suitability_trend_caption.pdf
```

Outputs The function writes a single-page PDF to:

```
<project_dir>/runs/<alpha_code>/Summaries/pages/<alpha_code>-Suitability-Trends.pdf
```

Layout and composition

- Two top-aligned maps (Suitability Trend and Change Trend) are placed side by side under a configurable top margin.
- The map row can be narrowed and centered using `map_width_factor`.
- A caption is placed `v_gap_in` inches below the taller of the two maps, centered horizontally and never stretched (aspect ratio preserved).
- Layout dimensions are controlled in inches and converted to pixels using the specified DPI.

Caption rendering The caption PDF (`suitability_trend_caption.pdf`) is rasterized via `pdfutils::pdf_convert()` at `caption_dpi` (default 600) to preserve original typography (including bold). It is trimmed to remove surrounding page whitespace, then centered horizontally and scaled down only if wider than the content area; otherwise it is drawn at native size.

Trimming

- Uniform borders are optionally trimmed from maps (controlled by `trim_maps`) to remove hidden padding.
- The caption is always trimmed after rasterization to remove the surrounding page whitespace, so only the text block contributes to the layout height calculation.
- Trimming uses a tolerance defined by `trim_fuzz`.

Methods Image reading, trimming, scaling, and composition are handled by the `magick` package. The caption PDF is rasterized by `pdfutils`. Aspect-preserving resizing ensures consistent layout while respecting maximum width and height constraints.

Data requirements All required input files must exist prior to execution. The function will terminate with an error if required PNGs or the caption PDF are missing.

Value

Character. Invisibly returns the output PDF file path.

Side effects:

- Writes a PDF file to the project directory.
- Creates output directories if they do not exist.
- Emits a console message indicating the output path.

Examples

```
## Not run:
assemble_suitability_trends_page("CASP")

## End(Not run)
```

```
assemble_variable_trends_page
  Assemble a "Variable Trends" page
```

Description

Builds a single-page PDF combining a variable-contribution lines plot, a summary table, and a caption into a structured layout.

Usage

```
assemble_variable_trends_page(
  alpha_code,
  page_width_in = 8.5,
  page_height_in = 11,
  dpi = 300,
  margin_left_in = 0.75,
  margin_right_in = 0.75,
  margin_bottom_in = 0.75,
  margin_top_in = 1,
  v_gap_in = 0.3,
  v_caption_gap_in = 0.3,
  plot_width_factor = 0.85,
  table_width_factor = 0.5,
  table_max_height_in = 3,
  table_pdf_dpi = 600L,
  trim_plot = TRUE,
  trim_table = TRUE,
  trim_fuzz = 8,
  caption_dpi = 600L,
  debug_frames = FALSE
)
```


Arguments

alpha_code	Character. Species alpha code (case-insensitive).
page_width_in	Numeric. Page width in inches. Default 8.5.
page_height_in	Numeric. Page height in inches. Default 11.
dpi	Numeric. DPI for raster composition. Default 300.
margin_left_in	Numeric. Left margin in inches. Default 0.75.
margin_right_in	Numeric. Right margin in inches. Default 0.75.
margin_bottom_in	Numeric. Bottom margin in inches. Default 0.75.
margin_top_in	Numeric. Top margin in inches. Default 1.00.
v_gap_in	Numeric. Vertical gap between plot and table in inches. Default 0.30.
v_caption_gap_in	Numeric. Vertical gap between table and caption in inches. Default 0.30.
plot_width_factor	Numeric. Fraction (0 to 1) of content width used by the plot. Default 0.85.
table_width_factor	Numeric. Fraction (0 to 1) of content width used by the table. Default 0.50.
table_max_height_in	Numeric. Maximum table height in inches. Default 3.00.
table_pdf_dpi	Integer. DPI used when rasterizing the table PDF. Default 600.
trim_plot	Logical. If TRUE, trim borders from the plot PNG. Default TRUE.
trim_table	Logical. If TRUE, trim borders from the table image. Default TRUE.
trim_fuzz	Integer. Tolerance (0 to 100 percent) for trimming. Default 8.
caption_dpi	Integer. DPI used when rasterizing the caption PDF. Default 600.
debug_frames	Logical. If TRUE, draw thin frames around placed images for debugging. Default FALSE.

Details

Pipeline context This function composes a precomputed variable-contribution trend plot, a summary table, and a package caption into a single standardized reporting page. It operates entirely within the rENM project directory and does not rely on external scripts.

Inputs Inputs must exist at the following locations:

```
<project_dir>/runs/<alpha_code>/Trends/variables/<alpha_code>-Variable-Contributions-BR-Lines-  
<project_dir>/runs/<alpha_code>/Summaries/tables/<alpha_code>-Variable-Trend-Summary.pdf
```

The caption is read from the installed package resource:

```
inst/captions/variable_trend_caption.pdf
```

Outputs The function writes a single-page PDF to:

```
<project_dir>/runs/<alpha_code>/Summaries/pages/<alpha_code>-Variable-Trends.pdf
```

Layout and composition

- The variable-contribution plot is placed at the top of the content area, scaled to fill the content width (subject to `plot_width_factor`) while preserving its aspect ratio.
- A summary table is placed `v_gap_in` inches below the plot, centered horizontally (subject to `table_width_factor`).
- A caption is placed `v_caption_gap_in` inches below the table, centered horizontally and never stretched (aspect ratio preserved).
- Layout dimensions are controlled in inches and converted to pixels using the specified DPI.

Caption rendering The caption PDF (`variable_trend_caption.pdf`) is rasterized via `pdftools::pdf_convert()` at `caption_dpi` (default 600) to preserve original typography (including bold). It is trimmed to remove surrounding page whitespace, then centered horizontally and scaled down only if wider than the content area; otherwise it is drawn at native size.

Table rendering The table PDF is rasterized via `pdftools::pdf_convert()` at `table_pdf_dpi` (default 600) to improve text clarity. The table is then fitted within `table_width_factor` of the content width subject to `table_max_height_in`.

Trimming

- Uniform borders are optionally trimmed from the plot (controlled by `trim_plot`) and the table (controlled by `trim_table`) to remove hidden padding.
- The caption is always trimmed after rasterization to remove the surrounding page whitespace, so only the text block contributes to the layout height calculation.
- Trimming uses a tolerance defined by `trim_fuzz`.

Methods Image reading, trimming, scaling, and composition are handled by the `magick` package. The table and caption PDFs are rasterized by `pdftools`. Aspect-preserving resizing ensures consistent layout while respecting maximum width and height constraints.

Data requirements All required input files must exist prior to execution. The function will terminate with an error if any required input or the caption PDF is missing.

Value

Character. Invisibly returns the output PDF file path.

Side effects:

- Writes a PDF file to the project directory.
- Creates output directories if they do not exist.
- Emits a console message indicating the output path.

Examples

```
## Not run:
assemble_variable_trends_page("CASP")

## End(Not run)
```

assemble_variable_trend_maps_page

Assemble "Variable Trend Maps" pages

Description

Builds a multi-page PDF that shows trends and acceleration/deceleration patterns of the top contributing variables.

Usage

```
assemble_variable_trend_maps_page(alpha_code, overwrite = TRUE, verbose = TRUE)
```

Arguments

alpha_code	Character. Species alpha code (e.g., "CASP"). Converted internally to uppercase for all paths and filenames.
overwrite	Logical. If TRUE (default) overwrites existing destination files; if FALSE the function stops if a destination file already exists.
verbose	Logical. If TRUE (default) prints processing steps and paths to the console.

Details

Pipeline context

- Copies a shared caption PDF (A) and a species-specific variable-trend-maps PDF (B) into the Summaries/pages directory for the target species.
- Ensures consistent naming so that downstream report builders pick up both pages automatically.

Inputs

- Shared caption PDF (A):
system.file("captions", "variable_trend_maps_caption.pdf", package = "rENM.reports")
- Species maps PDF (B):
file.path(rENM_project_dir(), "runs", <alpha_code>, "Summaries", "maps", paste0(<alpha_code>, "-Variable-Trend-Maps.pdf"))

Outputs

- Caption copy:
file.path(rENM_project_dir(), "runs", <alpha_code>, "Summaries", "pages", paste0(<alpha_code>, "-Variable-Trend-Maps1.pdf"))
- Maps copy:
file.path(rENM_project_dir(), "runs", <alpha_code>, "Summaries", "pages", paste0(<alpha_code>, "-Variable-Trend-Maps2.pdf"))
- Processing summary appended to:
file.path(rENM_project_dir(), "runs", <alpha_code>, "_log.txt")

Methods

- Pure file staging via file.copy() with optional overwrite.
- No PDF merging, rasterization, or resizing.

Value

A named list (invisibly returned):

- caption_page – full file path to the staged caption PDF
- maps_page – full file path to the staged maps PDF

Side effects:

- Two PDF files copied into <project>/runs/<alpha_code>/Summaries/pages
- Processing summary appended to <project>/runs/<alpha_code>/_log.txt

Examples

```
## Not run:
assemble_variable_trend_maps_page("CASP")

## End(Not run)
```

```
create_centroid_trend_summary_table
```

Create a centroid shift summary table

Description

Builds a two-part, publication-ready centroid summary for a species, combining a shift block and regression block into a standardized tabular format for reporting centroid trends. Exports to Excel and optionally to PNG and PDF.

Usage

```
create_centroid_trend_summary_table(
  alpha_code,
  decimals = 2,
  font_name = "Arial",
  title_size = 10,
  target_px = 350
)
```

Arguments

alpha_code	Character. Species alpha code (case-insensitive).
decimals	Integer. Number of decimals to display and format.
font_name	Character. Table font family.
title_size	Integer. Title font size in Excel and PNG/PDF.
target_px	Integer. Target pixel width used to center PNG tables.

Details

Pipeline context Builds a standardized centroid trend summary from precomputed centroid velocity and regression outputs for a given species.

Inputs Reads CSV files from:

- <project_dir>/runs/<ALPHA>/Trends/centroids/

Expected inputs:

Bioclimatic-Velocity <project_dir>/runs/<ALPHA>/Trends/centroids/ <ALPHA>-Bioclimatic-Velocity.csv
with columns start_lon, start_lat, end_lon, end_lat, distance_km, bearing_deg, velocity_km_per_yr.

Longitude-Summary <project_dir>/runs/<ALPHA>/Trends/centroids/ <ALPHA>-Centroids-Longitude-Summary.csv
with columns mean, ci_low, ci_high, pd, rope_pct.

Latitude-Summary <project_dir>/runs/<ALPHA>/Trends/centroids/ <ALPHA>-Centroids-Latitude-Summary.csv
with the same columns as Longitude-Summary.

Outputs Writes outputs to:

- <project_dir>/runs/<ALPHA>/Summaries/tables/ <ALPHA>-Centroid-Trend-Summary.xlsx
(always written)
- <project_dir>/runs/<ALPHA>/Summaries/tables/ <ALPHA>-Centroid-Trend-Summary.png
(if optional graphics packages are available)
- <project_dir>/runs/<ALPHA>/Summaries/tables/ <ALPHA>-Centroid-Trend-Summary.pdf
(if optional graphics packages are available)

Table structure

- Shift block (one row): 1980/2020 centroid coordinates, great-circle distance (km), bearing (deg), velocity (km/yr).
- Regression block (two rows: Longitude, Latitude): Slope, CI Low, CI High, PD, Rope \

Visual and typographic specification

- Title centered and bold; default 10 pt (configurable).
- Table text 8 pt (Arial default).
- No top border or rule on any block.
- No vertical lines anywhere.
- No horizontal rules between body rows in the regression block.
- Column-label rows have a light-gray ("BFBFBF") underline; each block capped with a light-gray bottom rule.
- PNG/PDF export uses **gt** plus **webshot2** or **chromote**, with both top and bottom tables centered using table.width and table.align and canvas padding via **magick**.
- PNG/PDF use 1 px data-row padding.

Logging Prints progress to the console and appends an eBird-style processing summary (72-dash ruler; Timestamp, Alpha code, Outputs saved, Output file, Total elapsed) to: <project_dir>/runs/<ALPHA>/_log.txt.

Value

Invisibly returns a List with:

- shift_df: Data frame containing the one-row shift block.
- reg_df: Data frame containing the two-row regression block.
- files: Character vector of file paths written.

Side effects:

- Writes Excel, PNG, and PDF files to the project directory.
- Appends a processing summary to the run-specific log file.

Examples

```
## Not run:
create_centroid_trend_summary_table("CASP")
create_centroid_trend_summary_table("GRRO",
  title_size = 11, font_name = "Helvetica")

## End(Not run)
```

```
create_suitability_trend_summary_table
  Create a state-level GAP range / hot spot summary table
```

Description

Produces a publication-ready summary table from state-level suitability trend data, exporting to Excel and optionally to image and PDF formats.

Usage

```
create_suitability_trend_summary_table(alpha_code)
```

Arguments

alpha_code Character. Four-letter species code.

Details

Pipeline context Converts state-level suitability trend summary outputs into formatted tables suitable for reporting, publication, and visualization.

Inputs Input CSV must exist at:

<project_dir>/runs/<alpha_code>/Trends/suitability/ <alpha_code>-Suitability-Trend-State-Analysis

Processing steps

- Reads the summary CSV using readr::read_csv.
- Verifies required columns are present.
- Converts numeric columns safely.

- Renames columns for presentation.
- Builds a formatted Excel workbook using openxlsx.
- Creates a styled table using gt.
- Optionally exports PNG using webshot2 if available.
- Optionally exports PDF using pagedown if available.

Outputs Output files are written to:

<project_dir>/runs/<alpha_code>/Summaries/tables

- Excel (.xlsx)
- PNG (.png) if webshot2 is available
- PDF (.pdf) if pagedown is available

Table columns are presented as: State, State Area, Range Area, Range \ Hot Spot Area, Hot Spot \

Log behavior Appends a processing summary to:

<project_dir>/runs/<alpha_code>/_log.txt

using the eBird-standard format.

Data requirements Input CSV must contain: state, state_area, range_area, range_pct, pos_pct, neg_pct, hotspot_area, hotspot_pct.

Value

A named list returned invisibly with the following elements:

- xlsx: Character. Absolute path to Excel output file.
- png: Character. Absolute path to PNG file or NA if skipped.
- pdf: Character. Absolute path to PDF file or NA if skipped.

Side effects:

- Writes formatted Excel, PNG, and PDF outputs.
- Appends a processing summary block to the run log file.

Examples

```
## Not run:
create_suitability_trend_summary_table("CASP")

## End(Not run)
```

```
create_variable_trend_summary_table
```

Create a variable trend summary table

Description

Produces a publication-ready summary table from variable-trend statistics with significance marks (*, **, ***) and bolding by PD.

Usage

```
create_variable_trend_summary_table(alpha_code)
```

Arguments

alpha_code Character. Species alpha code.

Details

Pipeline context Converts variable-level Bayesian trend statistics into a compact, formatted summary table suitable for reporting and publication.

Inputs Input file must exist at:

<project_dir>/runs/<alpha_code>/Trends/variables/ <alpha_code>-Variable-Contributions-BR-Stats.csv

Processing steps

- Reads the variable-trend CSV using readr::read_csv.
- Verifies required columns are present.
- Resolves CI column names: slope_ci_lower or slope_ci_low, and slope_ci_upper or slope_ci_high.
- Preserves row order exactly as in the input CSV.
- Assigns significance marks based on pd_slope:
 - *** if pd_slope >= 95
 - ** if 90 <= pd_slope < 95
 - * if 85 <= pd_slope < 90
- Appends significance marks to variable names.
- Converts numeric fields safely.
- Builds a formatted Excel workbook using openxlsx.
- Creates a styled table using gt.
- Optionally exports PNG using webshot2 if available.
- Optionally exports PDF using pagedown if available.

Outputs Output files are written to:

<project_dir>/runs/<alpha_code>/Summaries/tables

- Excel (.xlsx)
- PNG (.png) if webshot2 is available
- PDF (.pdf) if pagedown is available

Column mapping Output columns correspond to input columns:

- Variable <- Variable
- Points <- n_points
- Slope <- slope_mean
- CI Low <- slope_ci_lower or slope_ci_low
- CI High <- slope_ci_upper or slope_ci_high
- PD <- pd_slope
- ROPE % <- rope_slope_pct

Formatting rules

- Row order is preserved exactly as in the input CSV.
- Variables with pd_slope >= 85 are bolded.
- Significance marks are appended to variable names.

Log behavior Appends a processing summary to:

<project_dir>/runs/<alpha_code>/_log.txt

using the eBird-standard format:

- One blank line before the 72-dash separator.
- No ending separator line.
- Reports all output files written.

Data requirements Input CSV must contain: Variable, n_points, slope_mean, pd_slope, rope_slope_pct, and CI columns (slope_ci_lower or slope_ci_low, slope_ci_upper or slope_ci_high).

Value

A named list returned invisibly with the following elements:

- xlsx: Character. Absolute path to Excel output file.
- png: Character. Absolute path to PNG output or NA if skipped.
- pdf: Character. Absolute path to PDF output or NA if skipped.

Side effects:

- Writes Excel, PNG, and PDF files to disk.
- Appends a processing summary block to the run log file.

Examples

```
## Not run:
create_variable_trend_summary_table("CASP")

## End(Not run)
```

gather_range_maps	<i>Assemble 3x3 range map contact sheet</i>
-------------------	---

Description

Builds a 3x3 contact sheet of range maps for a species code by collecting PNG tiles from Time-Series outputs and arranging them on a standardized Letter-sized page layout for export to multiple formats.

Usage

```
gather_range_maps(
  alpha_code,
  margin_in = 1,
  page_width_in = 8.5,
  page_height_in = 11,
  tile = "3x3",
  trim_fuzz = 8,
  dpi = 300
)
```

Arguments

alpha_code	Character. Species code used in folder and file names.
margin_in	Numeric. Page margin on top, left, and right in inches.
page_width_in	Numeric. Page width in inches (Letter = 8.5).
page_height_in	Numeric. Page height in inches (Letter = 11).
tile	Character. Montage tile specification for magick::image_montage.
trim_fuzz	Numeric. Trim tolerance percent for magick::image_trim to remove light halos.
dpi	Numeric. Rendering resolution for raster-based layout.

Details

Pipeline context Assembles retrospective ENM range map outputs into a standardized multi-format visual summary for interpretation and reporting.

Inputs Source tiles are expected at:

<project_dir>/runs/<alpha_code>/TimeSeries/ <year>/model/ <alpha_code>-<year>-Range.png
for year in 1980, 1985, ..., 2020 (9 images total).

Processing steps

- Reads and trims each tile to remove stray white borders using magick::image_trim.
- Creates a 3x3 montage with small gutters.
- Trims the montage again to tighten outer bounds.
- Scales the montage to the content width (6.5 inches at 300 dpi).
- If the montage exceeds the content height (9 inches at 300 dpi), crops excess from the bottom while preserving the top.
- Composites onto a white Letter-sized page.

Outputs The final page is exported as:

- A one-page PNG (<alpha_code>-Range-Maps.png)
- A one-page PDF (<alpha_code>-Range-Maps.pdf)
- A one-page DOCX (<alpha_code>-Range-Maps.docx)

All files are written to:

<project_dir>/runs/<alpha_code>/Summaries/maps

The DOCX placement matches the PNG/PDF visually: the montage is scaled to the page content width (page width minus 2 x margin) and cropped from the bottom if it exceeds the content height, preserving the top of the montage so it sits 1 inch from the top margin and is centered horizontally.

Data requirements Exactly 9 range PNG files must exist for the specified alpha_code.

Value

A named list returned invisibly with the following elements:

- png: Character. Absolute file path to PNG output.
- pdf: Character. Absolute file path to PDF output.
- docx: Character. Absolute file path to DOCX output.

Side effects:

- Writes PNG, PDF, and DOCX files to the summaries directory.
- Appends a processing summary to: <project_dir>/runs/<alpha_code>/_log.txt

Examples

```
## Not run:
# For BETH using default layout and margins:
gather_range_maps("BETH")

# For a different alpha_code:
gather_range_maps("AMRO")

# Tighter trim if input PNGs have larger white borders:
gather_range_maps("BETH", trim_fuzz = 12)

## End(Not run)
```

gather_suitability_maps

Assemble side-by-side suitability maps into paged composites

Description

Builds a 3x3 contact sheet of suitability maps for a species code by collecting prediction PNGs from TimeSeries outputs and arranging them on a standardized Letter-sized page layout for export to multiple formats.

Usage

```
gather_suitability_maps(
  alpha_code,
  margin_in = 1,
  page_width_in = 8.5,
  page_height_in = 11,
  tile = "3x3",
  trim_fuzz = 8,
  dpi = 300
)
```

Arguments

alpha_code	Character. Species code used in folder and file names.
margin_in	Numeric. Page margin on top, left, and right in inches.
page_width_in	Numeric. Page width in inches (Letter = 8.5).
page_height_in	Numeric. Page height in inches (Letter = 11).
tile	Character. Montage tile specification for magick::image_montage.
trim_fuzz	Numeric. Trim tolerance percent for magick::image_trim to remove light halos.
dpi	Numeric. Rendering resolution for raster-based layout.

Details

Pipeline context Assembles retrospective ENM prediction outputs into a standardized multi-format visual summary for interpretation and reporting.

Inputs Source tiles are expected at:

<project_dir>/runs/<alpha_code>/TimeSeries/ <year>/model/ <alpha_code>-<year>-Prediction.png
for year in 1980, 1985, ..., 2020 (9 images total).

Processing steps

- Reads and trims each tile to remove stray white borders using magick::image_trim.
- Creates a 3x3 montage with small gutters.
- Trims the montage again to tighten outer bounds.
- Scales the montage to the content width (6.5 inches at 300 dpi).
- If the montage exceeds the content height (9 inches at 300 dpi), crops excess from the bottom while preserving the top.
- Composites onto a white Letter-sized page.

Outputs The final page is exported as:

- A one-page PNG (<alpha_code>-Suitability-Maps.png)
- A one-page PDF (<alpha_code>-Suitability-Maps.pdf)
- A one-page DOCX (<alpha_code>-Suitability-Maps.docx)

All files are written to:

<project_dir>/runs/<alpha_code>/Summaries/maps

The DOCX placement matches the PNG/PDF visually: the montage is scaled to the page content width (page width minus 2 x margin) and cropped from the bottom if it exceeds the content height, preserving the top of the montage so it sits 1 inch from the top margin and centered horizontally.

Data requirements Exactly 9 prediction PNG files must exist for the specified alpha_code.

Value

A named list returned invisibly with the following elements:

- png: Character. Absolute file path to PNG output.
- pdf: Character. Absolute file path to PDF output.
- docx: Character. Absolute file path to DOCX output.

Side effects:

- Writes PNG, PDF, and DOCX files to the summaries directory.
- Appends a processing summary to: <project_dir>/runs/<alpha_code>/_log.txt

Examples

```
## Not run:
# For BETH using default layout and margins:
gather_suitability_maps("BETH")

# For a different alpha_code:
gather_suitability_maps("AMRO")

# Tighter trim if input PNGs have larger white borders:
gather_suitability_maps("BETH", trim_fuzz = 12)

## End(Not run)
```

```
gather_suitability_trend_stats
```

Merge state-level suitability and hot-spot statistics

Description

Reads two per-state CSV files for a species and merges them by two-letter state abbreviation to produce a concise summary table of suitability and hotspot metrics.

Usage

```
gather_suitability_trend_stats(alpha_code)
```

Arguments

alpha_code Character. Four-letter species code.

Details

Pipeline context Combines state-level suitability trend outputs with hotspot statistics into a unified summary table for downstream reporting and analysis.

Inputs Input files must exist at:

<project_dir>/runs/<alpha_code>/Trends/suitability/ <alpha_code>-Suitability-Trend-State-Analysis

<project_dir>/runs/<alpha_code>/Trends/suitability/ <alpha_code>-Suitability-Trend-State-Analysis

Processing steps

- Reads both input CSV files using `readr::read_csv`.
- Verifies required columns are present in each dataset.
- Sorts both inputs alphabetically by state abbreviation.
- Merges datasets using a left join on state abbreviation.
- Converts relevant columns to numeric values.
- Assembles a standardized summary table with key metrics.

Outputs Output file is written to:

`<project_dir>/runs/<alpha_code>/Trends/suitability/ <alpha_code>-Suitability-Trend-State-Analysis`

Log behavior Appends a processing summary to:

`<project_dir>/runs/<alpha_code>/_log.txt`

using the eBird-standard format:

- Always skips one blank line before the 72-dash separator.
- Does not include an ending separator line.
- Omits all raster-related fields.
- Includes: Timestamp, Alpha code, Outputs saved, Total elapsed, and Output file.

Data requirements

- First file must contain: STATE, GAP.RANGE.AREA, GAP.RANGE.PCT, GAP.RANGE.POS.PCT, and either GAPP.RANGE.NEG.PCT or GAP.RANGE.NEG.PCT.
- Second file must contain: abbr, state_area_km2, hotspot_area_km2, hotspot_pct_of_state.

Value

Invisibly returns a `data.frame` with the following columns:

- `state`: Character. Two-letter state abbreviation.
- `state_area`: Numeric. State area in square kilometers.
- `range_area`: Numeric. Area of species range.
- `range_pct`: Numeric. Percent of state occupied by range.
- `pos_pct`: Numeric. Percent of positive trend area.
- `neg_pct`: Numeric. Percent of negative trend area.
- `hotspot_area`: Numeric. Area classified as hotspot.
- `hotspot_pct`: Numeric. Percent of state as hotspot.

Side effects:

- Writes a summary CSV file to the suitability directory.
- Appends a processing summary block to the run log file.

Examples

```
## Not run:
gather_suitability_trend_stats("CASP")

## End(Not run)
```

gather_top_variable_trend_maps

Assemble side-by-side variable trend maps into paged composites

Description

Pair trend and rate PNGs for each variable, build A | B montages, arrange several montages per page, and export them as PDF, DOCX, and PNG. Designed for variable-contribution reporting in the rENM workflow.

Usage

```
gather_top_variable_trend_maps(
  alpha_code,
  base_dir = c("auto", "m2", "mc"),
  base_home = "~",
  vars_csv_subpath = NULL,
  per_page = 5,
  dpi = 300,
  page_width_in = 8.5,
  page_height_in = 11,
  margin_in = 1,
  trim_fuzz = 8
)
```

Arguments

alpha_code	Character. Species code (e.g., "CASP"). Used for input and output file naming.
base_dir	Character. One of "auto", "m2", "mc"; controls which MERRA-2 directories are searched for trend images.
base_home	Character. Deprecated. Ignored and retained only for backward compatibility; the project root is taken from <code>rENM_project_dir()</code> .
vars_csv_subpath	Character or NULL. Optional path (relative to <code>runs/<alpha_code></code>) to the variable list CSV; when NULL the default path described in Pipeline steps is used.
per_page	Integer. Number of variables (rows) per output page.
dpi	Numeric. Rendering resolution in dots per inch.
page_width_in	Numeric. Page width in inches.
page_height_in	Numeric. Page height in inches.
margin_in	Numeric. Page margin in inches on all sides.
trim_fuzz	Numeric. Fuzz parameter passed to <code>magick::image_trim()</code> (in pixels).

Details

Pipeline steps

- 1. Read the variable list from `<rENM_project_dir()>/runs/<alpha_code>/Trends/variables/<alpha_code>-Variable-Contributions-BR-Stats.csv` (must contain Variable; `pd_slope` is optional).

- 2. For every variable <var> collect two PNGs (case-insensitive, punctuation-agnostic matching):
 - A (trend): <base>-trends/<var>.png
 - B (rate) : <base>-trends-rate/<var>.png

If base_dir == "auto" both "m2" and "mc" directories are tried.

- 3. Build an A | B montage for each variable with generous spacing.
- 4. Stack per_page montages per page.
- 5. Trim, down-scale, and vertically crop overflow.
- 6. Composite onto a white US Letter page (8.5 x 11 in, 1 in margins).
- 7. Export a multipage PDF, a matching DOCX, and page-level PNGs named -p01, -p02, ...

Highlight rule If the CSV includes pd_slope, any variable with {pd_slope >= 90} receives a dark-blue strip along the left edge.

Layout and styling

- Each A <-> B pair is slightly down-scaled to add whitespace.
- The gutter between A and B, as well as row spacing, is widened.
- The full column is scaled to 96\
- Excess height beyond the content area is cropped from the bottom.

CRAN compliance All hard-coded paths have been removed; the project root is obtained via rENM_project_dir().

Inputs

- Variable CSV (with optional pd_slope) as described above.
- Trend and rate PNGs residing in the two <base>-trends* folders under <rENM_project_dir()>/data/merra/.

Outputs

- <alpha_code>-Variable-Trend-Maps.pdf
- <alpha_code>-Variable-Trend-Maps.docx
- <alpha_code>-Variable-Trend-Maps-pXX.png
- A processing summary appended to runs/<alpha_code>/_log.txt

Value

A list with the following elements:

- png: Character vector of page-level PNG paths.
- pdf: Character string giving the PDF path.
- docx: Character string giving the DOCX path.

Side effects:

- Files are written to runs/<alpha_code>/Summaries/maps/.
- A log block is appended to runs/<alpha_code>/_log.txt.

Examples

```
## Not run:
gather_top_variable_trend_maps("CASP")          # auto-detect bases
gather_top_variable_trend_maps("CASP", "m2")     # force "m2"
gather_top_variable_trend_maps("CASP", per_page = 3)

## End(Not run)
```

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